# Predicting Oral Health Outcomes from Socioeconomic and Access-to-Care Features in U.S. Adults: A Statistical Learning Analysis of the 2017–2018 NHANES Cycle

Jieyu Deng Bryce Grover Juncheng Han Jeffrey Pinarchick

2026-05-01

Oral diseases affect billions of people worldwide and are among the most common chronic health conditions. In the United States, oral health outcomes, access to dental care, and treatment needs are strongly associated with socioeconomic factors such as income, education, and race. Using data from the 2017 to 2018 National Health and Nutrition Examination Survey, we constructed a dataset of 5,265 adults aged 20 and older that combined demographic, socioeconomic, behavioral, and clinical dental information. We compared supervised statistical learning methods across three tasks: predicting self-rated fair or poor oral health, predicting whether an examiner recommended dental care, and predicting the continuous Decayed, Missing, and Filled Teeth (DMFT) score using only socioeconomic and access-related variables. Across all three tasks, gradient-boosted trees outperformed regularized linear models and random forests. The best model achieved a cross-validated ROC-AUC of 0.849 for self-rated poor oral health and 0.844 for examiner-identified need for care. For DMFT, a two-stage regression approach that first identified edentulous adults and then estimated DMFT among dentate adults reduced mean absolute error from 6.98 to 4.67 teeth. Feature importance analysis showed that income-to-poverty ratio, age, time since last dental visit, and cost-related barriers to care were the strongest predictors across tasks. These results suggest that major oral health disparities among U.S. adults can be predicted using a relatively small set of commonly collected socioeconomic variables.

## Introduction and Literature Review

Oral diseases affect an estimated 3.7 billion people globally and untreated dental caries in permanent teeth ranks as the world's most common health condition (World Health Organization 2025; Peres et al. 2019). The personal consequences of chronic untreated disease (pain, sepsis, functional limitation, and reduced quality of life) fall disproportionately on people living in poverty and on socially marginalized groups, a pattern that holds across high, middle, and low-income countries alike (Peres et al. 2019). In the United States this global pattern is especially salient. By age 50, the typical U.S. adult has experienced caries in about two-thirds of their permanent teeth, and

nearly one in nine adults aged 20 and older has lost all of their natural teeth (Dye, Weatherspoon, and Lopez Mitnik 2019). The direct costs of dental disease are considerable. The U.S. spends more than $140 billion per year on dental services. Unlike many chronic conditions, oral disease is very preventable through a combination of behavioral, clinical, and structural interventions whose effectiveness is well established. The persistence of disease is less often a question of biomedical reasons and more often one of who has access to dental care and who does not.

The socioeconomic down-slope in oral health is among the most established findings in public-health research. Adults below the federal poverty line have roughly twice the rate of untreated decay of adults earning five or more times the poverty threshold. This gap has been stable for two decades (Braveman, Egerter, and Williams 2011; Sabbah et al. 2007). Similar disparities appear across education, insurance coverage, and race and ethnicity. Self-rated oral health, which measures patients' own perception of their dental status, shows comparable patterns and is an independent predictor of functional impact, quality of life, and health-care utilization (Locker and Quinonez 2009; Slade 1997).

The literature on modeling oral-health outcomes from survey data has generally used logistic regression with a small number of pre-selected covariates. These analyses emphasize interpretable coefficient estimates suitable for determining whether a specific exposure causes a particular health outcome but underutilize two capabilities of modern statistical learning. The first is the ability to discover non-linearity and interactions among many features, and the second, the ability to quantify out-of-sample predictive performance rather than in-sample model fit alone (James et al. 2021). Both capabilities matter for policy-facing questions.

This project presents a comparative statistical learning analysis of three oral-health outcomes using a nationally representative sample of U.S. adults. It addresses three related questions. First, how accurately can an adult's self-rated oral health be predicted using demographic characteristics, socioeconomic status, and reported access to and use of dental care? Second, how accurately can a clinician's recommendation for dental treatment be predicted using those same variables along with a small set of self-reported oral-health symptoms? Third, how much variation in the continuous Decayed-Missing-Filled-Teeth (DMFT) score, a common overall measure of lifetime dental decay, can be explained using only upstream socioeconomic and access-related predictors, without direct clinical examination data?

To answer these questions, we use both shallow and deep learning approaches. For the three final prediction tasks, we compare a regularized linear model, a random forest, and gradient-boosted trees (XGBoost). Earlier in the project, we also developed a deep learning track based on feedforward neural networks that examined the relationship between dental health and income from a different direction, using the DMFT score to study income-related outcomes. Although that exploratory work is not directly aligned with the final tasks reported here, it helped shape the final study design and informed the comparison among the shallow methods. We also account for an important feature of the DMFT outcome: many adults have no remaining natural teeth, which creates a large cluster of values at the top of the scale. To address this, we use a two-stage regression framework that first predicts complete tooth loss and then estimates disease burden among those who still have natural teeth.

The primary goal of this project is prediction rather than causal inference. We do not claim that factors such as income or education directly cause poor oral-health outcomes. Instead, we examine how well these well-established associations translate into out-of-sample predictive performance

when practical challenges in survey data are properly addressed, including data leakage, class imbalance, missingness, and complex sampling design.

## Data Collection

### Source and scope

All data come from the 2017–2018 cycle of the National Health and Nutrition Examination Survey (NHANES), a continuous, nationally representative, cross-sectional survey administered by the National Center for Health Statistics (Centers for Disease Control and Prevention, National Center for Health Statistics 2020). Each participating adult completes an in-home interview, a Mobile Examination Center (MEC) clinical exam, and a self-administered questionnaire. NHANES releases the corresponding datasets as separate SAS transport files keyed on a common sequence number (`SEQN`). We downloaded four files directly from the public NHANES distribution:

- **DEMO_J** (demographics): age, sex, race and ethnicity, education, marital status, household composition, household income, and the income-to-poverty ratio (IPR).
- **OHXDEN_J** (oral-health dentition exam): tooth-by-tooth status and caries coding for all 32 tooth positions, plus summary flags for implants and root caries.
- **OHQ_J** (oral-health questionnaire): self-rated oral health, time since last dental visit, reasons for visits, reported unmet care, and 11 specific barriers-to-care indicators.
- **OHXREF_J** (oral-health recommendation of care): the examiner's overall recommendation code and diagnosis-specific referral flags.

The four files were merged on `SEQN` with a left join against `DEMO_J` as the anchor.

### Eligibility and filtering

We restricted the merged dataset to adults aged 20 and older who completed both the interview and the MEC exam (NHANES variable `RIDSTATR == 2`). Adults 20+ were chosen because their dentition is stable (they have all of their adult teeth) and because the access-to-care and self-rated health items on the questionnaire are asked only of adult respondents. Applying both filters gives a working dataset of 5,265 adult participants.

### Target variables

We derived three targets from the merged file.

*Target 1 — Poor self-rated oral health* (`target_poor_selfrated`, binary) was coded 1 for participants whose answer to the NHANES self-rated oral-health item (`OHQ845`) was *Fair* (4) or *Poor* (5) and 0 otherwise. The positive rate in the full sample is 34.4%.

*Target 2 — Clinician recommendation for care* (`target_needs_care`, binary) was coded 1 when the NHANES examiner issued any care recommendation other than "continue routine care" (`OHAREC`

< 4). The observed positive rate is 42.5% among the 5,057 participants with a non-missing recommendation. The remaining 208 participants had a missing recommendation code and were excluded from Task 2 but kept elsewhere.

*Target 3 — DMFT score* (`dmft_score`, continuous 0–28) was computed from the 28 caries codes (`OHXnnCTC`) in the dentition exam, summing decayed, filled-decayed, filled-sound, and missing-due-to-caries tooth counts in the classical Klein-Palmer method (Klein, Palmer, and Knutson 1938). There is a pronounced right-tail at 28 corresponding to edentulous adults (no natural teeth).

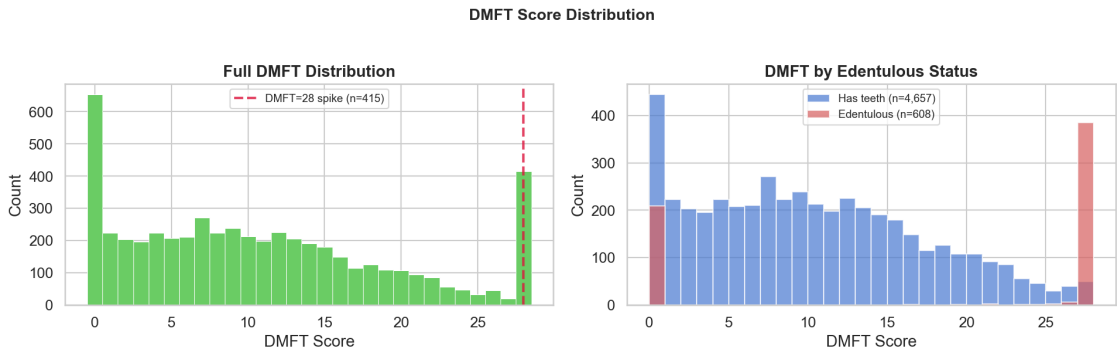


Figure 1: DMFT score distribution. The left panel shows the full sample. The right panel separates edentulous adults (n = 608) from those with at least one natural tooth (n = 4,657). The structural spike at 28 motivated the two-stage modeling approach.

For convenience, Table 1 summarizes the three prediction tasks referenced throughout the rest of the paper.

Table 1: Quick reference for the three prediction tasks evaluated in this paper. Task 2 excludes 208 participants with a missing examiner recommendation; Tasks 1 and 3 use the full sample.

| Task | Outcome (NHANES variable) | Type | Sample size | Positive rate / mean |
|---|---|---|---|---|
| **Task 1** | Poor self-rated oral health (`target_poor_selfrated`, derived from `OHQ845`) | Binary classification (1 = *Fair* or *Poor*) | 5,265 | 34.4% positive |
| **Task 2** | Clinician recommendation for any care (`target_needs_care`, derived from `OHAREC`) | Binary classification (1 = recommendation other than "continue routine care") | 5,057 | 42.5% positive |
| **Task 3** | Decayed-Missing-Filled-Teeth score (`dmft_score`) | Continuous regression (0–28) | 5,265 | mean 10.74 (SD 8.42) |

## Feature engineering and cleaning

Raw clinical tooth-level exam codes are not well suited for modeling, so we converted them into a smaller set of interpretable summary features. Specifically, we collapsed the 28 `OHXnnCTC` columns and 32 `OHXnnTC` columns into counts of decayed, filled-sound, filled-decayed, missing-due-to-caries, sound, and implanted teeth, along with the number of teeth present, the DMFT score, a treatment

ratio, and a binary `is_edentulous` indicator for participants with no remaining natural teeth (`n = 608`, 11.5%). The treatment ratio was defined as the number of filled-sound teeth divided by the number of teeth that had ever experienced decay, and it served as a proxy for access to restorative care.

Several NHANES-specific coding conventions also had to be addressed before these variables could be used in a model. In NHANES, refusals and "don't know" responses are stored as numeric values such as `7` or `77` for *Refused* and `9` or `99` for *Don't know*. Because these codes are numeric, they can be mistakenly treated as meaningful values by statistical models. We therefore replaced them with `NaN` in eleven questionnaire variables.

NHANES also codes yes/no items as `1 = Yes` and `2 = No`, so we re-coded five such variables to the more standard `{0, 1}` format. This change was especially important for interpreting logistic regression coefficients. We also adjusted the income-to-poverty ratio by clipping a small number of near-zero floating-point artifacts and enforcing the survey's upper limit of `5.0`, resulting in a final range of $[0.05, 5.0]$. Finally, the eleven care-barrier items were converted into binary indicators. Because these variables are only populated for adults who reported any unmet dental need, missing values in those columns were interpreted as indicating that the specific barrier was not reported.

## Exploratory Data Analysis

### Sample Characteristics and Target Distributions

Before building any models, we first looked at how the working sample of 5,265 adults is distributed across the three prediction targets. Figure 2 shows these distributions. Neither of the two yes-or-no outcomes is severely lopsided. About 34.4% of respondents rated their oral health as fair or poor, and 42.5% of those with a recorded examiner code were told they needed some form of care beyond routine maintenance. Because both groups are reasonably well represented, the evaluation tools used later in the paper are expected to behave reliably. The DMFT target has an average value of 10.74 teeth, a spread (standard deviation) of 8.42, and covers the full possible range from 0 to 28. A noticeable cluster at the upper end of the scale is already visible, and it corresponds to adults with no remaining natural teeth, a group that will receive special treatment in the modeling section.

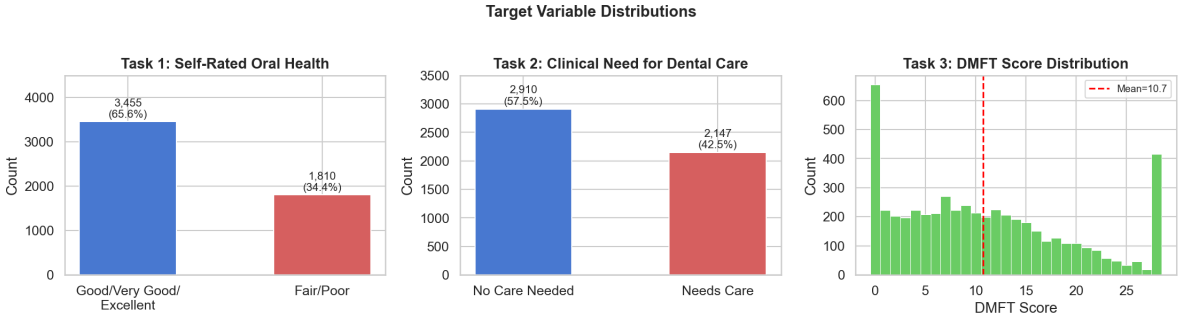


Figure 2: Distributions of the three prediction targets. Task 1 (*target_poor_selfrated*) and Task 2 (*target_needs_care*) show moderate class imbalance. Task 3 (*dmft_score*) shows a large cluster of high values corresponding to adults with no remaining natural teeth.

The demographic composition of the sample (Figure 3) reflects the way NHANES is designed. The mean age is 51.4 years (standard deviation of 17.7), about 51.7% of participants are female, and the self-identified race and ethnicity breakdown is 34.3% Non-Hispanic White, 23.6% Non-Hispanic Black, 19.4% in the combined Non-Hispanic Asian or Other category, 13.3% Mexican American, and 9.4% Other Hispanic. NHANES deliberately includes more participants from minority groups than a simple random sample would, in order to produce more stable estimates for those groups (Centers for Disease Control and Prevention, National Center for Health Statistics 2020), and this design choice is clearly visible in the figure.

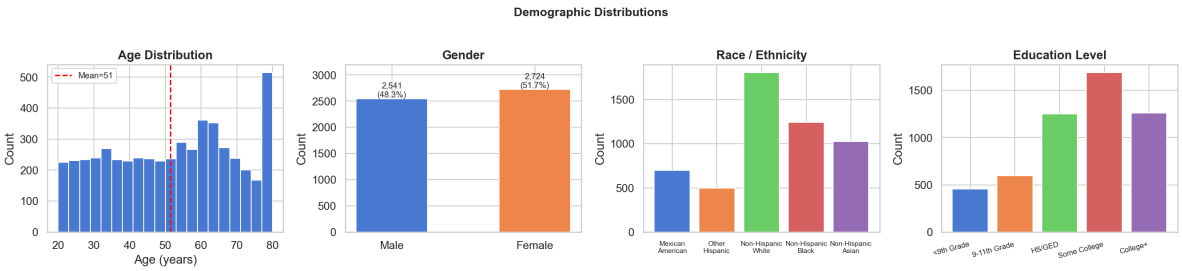


Figure 3: Demographic distributions in the working sample of 5,265 adults: age, sex, and self-identified race and ethnicity. The race and ethnicity breakdown reflects NHANES oversampling of minority populations.

## Socioeconomic and Access-to-Care Patterns

The distribution of the income-to-poverty ratio (Figure 4) has two features worth noting. The lower end is heavy, with 16.5% of adults living below the federal poverty line, and a visible spike appears at the value of 5.0 because NHANES caps the ratio at five times the poverty threshold. Any household earning above that limit is recorded at exactly 5.0. When the income-to-poverty ratio is compared across levels of self-rated oral health, a clear and steady pattern emerges. Adults who rate their oral health as poor have noticeably lower mean incomes than adults who rate it as excellent, and a similar pattern holds across levels of education. These simple comparisons already provide descriptive support for the social gradients in oral health (Braveman, Egerter, and Williams 2011; Sabbah et al. 2007) discussed in the introduction.

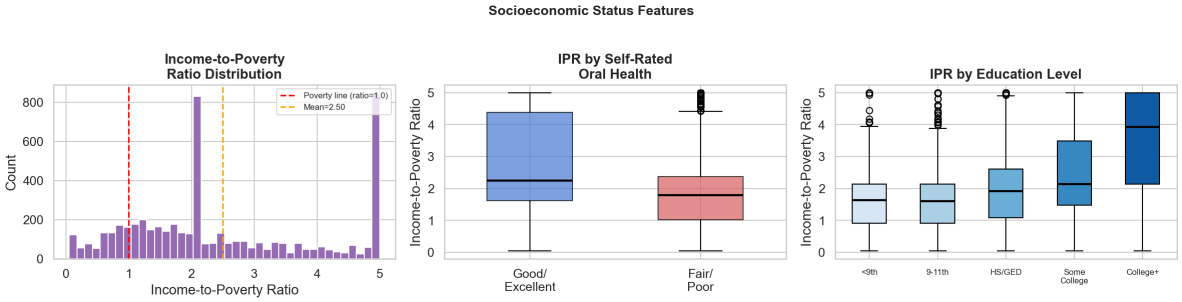


Figure 4: Socioeconomic features in the working sample. Left: income-to-poverty ratio distribution with a spike at the survey cap of 5.0. Middle and right: income-to-poverty ratio compared across self-rated oral health and education, showing consistent gradients.

Access-to-care variables (Figure 5) tell a consistent story. Time since the last dental visit is strongly skewed, with a substantial portion of the sample reporting more than a year since their previous visit. Among the eleven specific barriers to care, cost stands out: about 15.1% of adults cite it as a reason for not obtaining needed dental care, far more than cite insurance limitations or scheduling problems. This pattern supports the view that financial access, rather than informational or attitudinal factors, is the main obstacle to care in this population (Northridge, Kumar, and Kaur 2020).

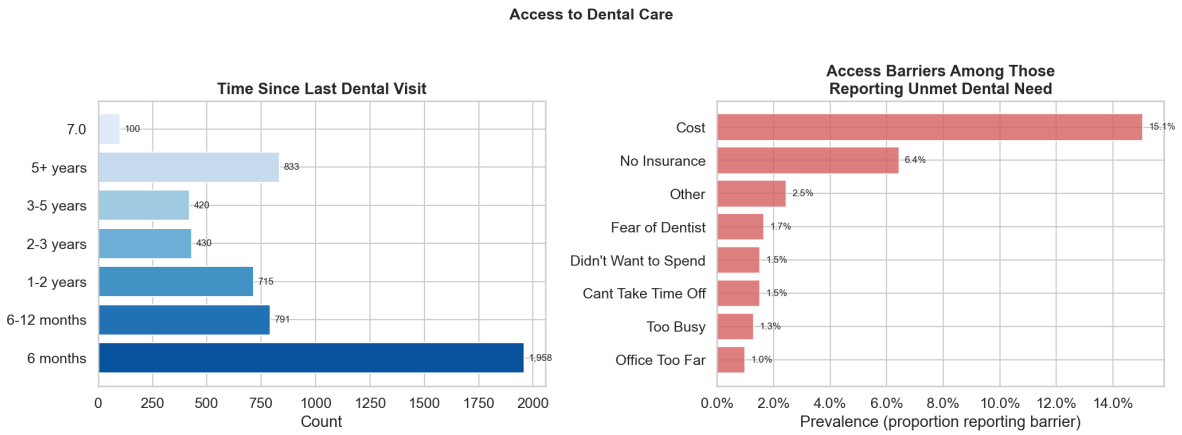


Figure 5: Access-to-care distributions. Left: time since last dental visit, strongly skewed toward long gaps. Right: prevalence of the eleven reported barriers to care, with cost dominant at 15.1%.

## Clinical Feature Distributions

The engineered clinical features (Figure 6) have distinctive shapes that influence modeling choices later on. The distribution of the number of teeth present has one main peak near the full complement of 28, a second, smaller peak at 0 representing adults with no remaining teeth, and a long tail covering adults with partial tooth loss. The count of untreated decayed teeth is heavily skewed to the right because 82.7% of adults have zero currently decayed teeth. This is consistent with a treated population in which fillings have largely kept pace with new decay. The treatment ratio, defined earlier as the share of previously decayed teeth that are now filled, has two distinct peaks, one near zero and one near one, which reflects a clear separation between adults whose dental disease has been well managed over time and those whose has not.

Breaking DMFT out by age quartile, education, and income (Figure 9) makes the effect of lifetime accumulation easy to see. The median DMFT score is roughly 3 in the youngest age quartile but climbs to about 19 in the oldest. The gradients by education and income point in the same direction, although their slopes are less steep. These descriptive patterns foreshadow the feature-importance rankings that appear later in the paper.

## Correlation Structure and Missingness

The missing-data patterns in the working dataset (Figure 7) are largely structural rather than the result of random non-response. A cluster of oral-health questionnaire items shows missingness around 14.7%, and this cluster exists because those items are only asked of respondents who first

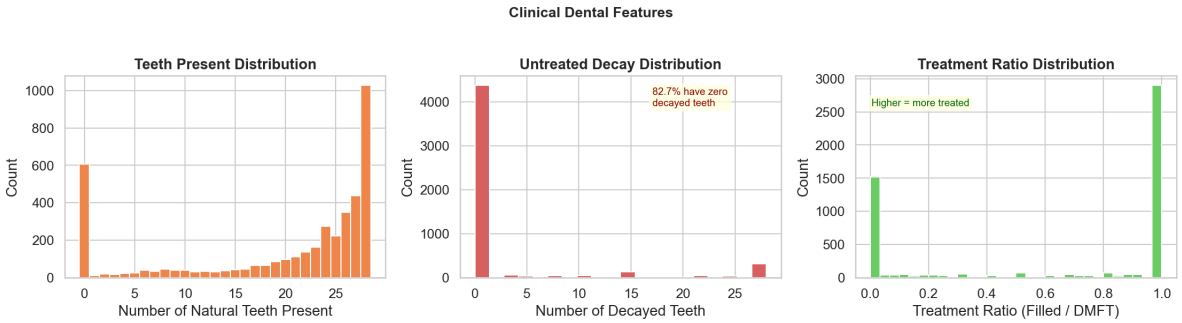


Figure 6: Clinical summary feature distributions: teeth present (with a secondary peak at 0), untreated decayed-tooth counts (strongly right-skewed), and the treatment ratio (with two distinct peaks).

indicate that they needed dental care but did not receive it. As noted in the Data Collection section, treating the absent values as indicating that a given barrier was not reported is consistent with the way the survey was designed, and it avoids inventing variability through imputation.

The correlation heatmap (Figure 8) confirms several expected relationships among the continuous variables. A correlation (r) measures how closely two variables move together on a scale from -1 to 1, where values near zero indicate little linear relationship. DMFT is positively related to age (r = 0.60), meaning that DMFT tends to rise as age rises, and negatively related to the number of teeth present (r = -0.61), since tooth loss naturally reduces the set of teeth available to be decayed, filled, or missing. The income-to-poverty ratio is moderately negatively related to both DMFT and to reports of cost as a barrier. The strongest correlation visible in the heatmap is between the underlying self-rated oral health item and the Task 1 target (r = 0.82). This very high value is expected rather than informative, because the Task 1 target is derived directly from that item, and we include it here as a sanity check. To avoid using an outcome to predict itself, the self-rated oral health item is excluded from the feature set for Task 1, as described in the Methods section.

# Methods

## Feature sets

Each of the three modeling tasks used a slightly different collection of predictor variables. Task 1 used 32 features, Task 2 used 36, and Task 3 used 28. Across tasks, these predictors fell into five broad groups: socioeconomic background, demographics, access to and use of dental care, oral-health-related behaviors, and, for Tasks 2 and 3, a small set of self-reported symptoms describing how the participant experienced their own oral health.

The socioeconomic variables included measures such as income relative to the poverty line and educational attainment. The demographic variables included age, sex, race and ethnicity, marital status, household size, and country of birth. The access variables captured factors such as time since the last dental visit, whether dental care was needed but not obtained, and the eleven reported barriers to receiving care described in the Data Collection section.

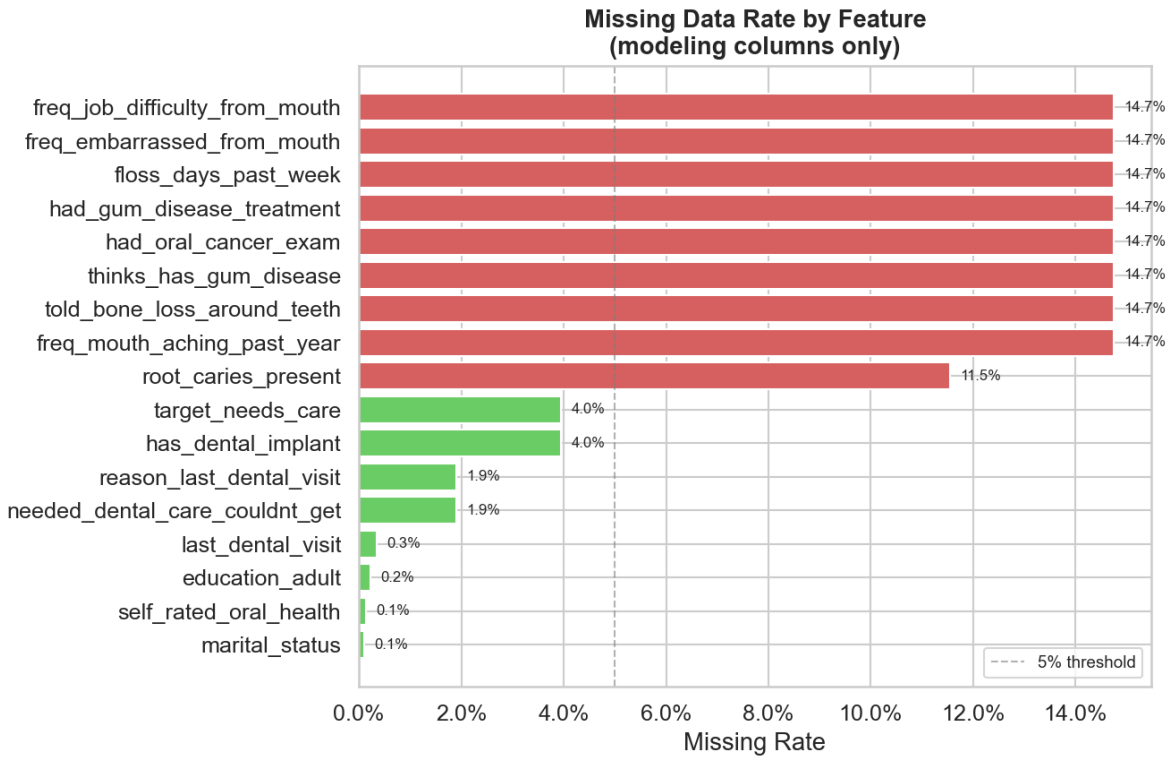


Figure 7: Missing-data rates across variables in the merged file. The roughly 14.7% missingness among oral-health questionnaire barrier items reflects the NHANES skip pattern rather than random non-response.

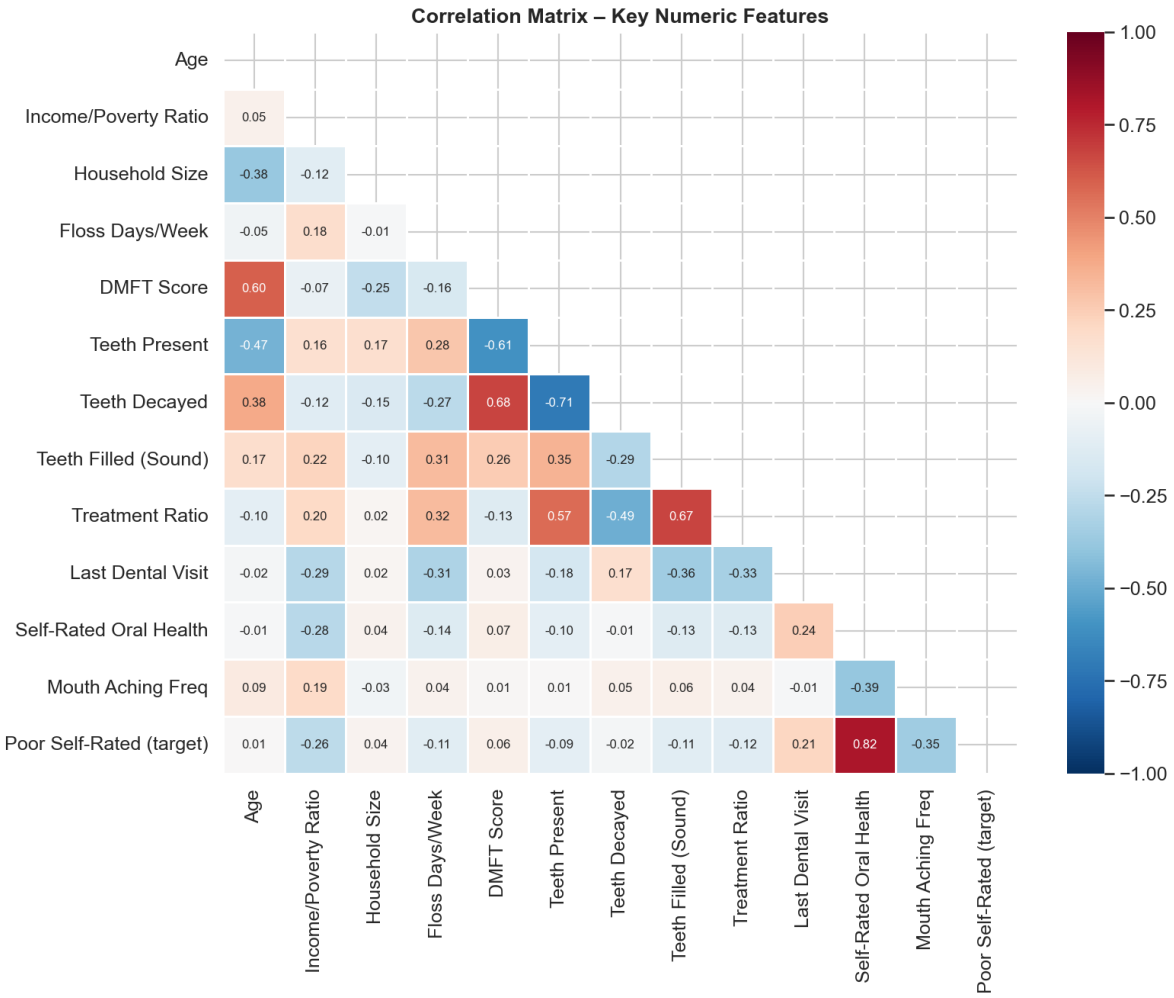


Figure 8: Correlation heatmap among continuous predictors and outcomes. Age-DMFT (r = 0.60) and DMFT-teeth present (r = -0.61) are the strongest non-trivial associations. The r = 0.82 correlation between *self_rated_oral_health* and the Task 1 target reflects the target's derivation from that item and is handled through feature exclusion.

The behavioral variables included flossing frequency, prior treatment for gum disease, and prior oral cancer examination. For Tasks 2 and 3, we also included a small set of self-reported symptom variables to reflect how participants described their own oral-health problems.

## Models and cross-validation

For each of the three prediction tasks, we compared three model families, moving from a simple linear baseline to more flexible non-linear methods based on decision trees. We also include, as a fourth modeling component, an earlier deep learning analysis that helped shape the final framing of the project.

1. **Logistic regression (Tasks 1 and 2) or Ridge regression (Task 3).** These models served as our baseline. For the two classification tasks, we used logistic regression with standardized inputs and balanced class weights to account for the moderately unequal positive rates. For the regression task, we used ridge regression with $\alpha = 1.0$. These linear models are easy to interpret and provide a clear benchmark for the more flexible methods that follow.

2. **Random forest.** A random forest combines many decision trees, each trained on a random subsample of the data and restricted to a random subset of features at each split (Breiman 2001). We used 300 to 500 trees, the usual square-root feature-sampling rule, and balanced class weights for the classification tasks. Random forests handle non-linear relationships well and are relatively unaffected by correlated predictors.

3. **Gradient-boosted trees (XGBoost).** XGBoost builds its trees sequentially rather than independently, with each new tree focused on the errors made by the previous ones. We used 300 to 500 trees, a learning rate of 0.05, and a maximum depth of 4. For the classification tasks, we set the positive-class weight to the ratio of negative to positive cases to handle imbalance. For Tasks 1 and 2, we ran a randomized hyperparameter search over the number of trees, maximum depth, learning rate, subsample ratio, column-subsample ratio, minimum child weight, and gamma, using 5-fold stratified cross-validation with ROC-AUC as the tuning metric.

4. **Feedforward neural networks (deep learning).** Earlier in the project, we also explored a separate deep learning track that addressed a different question. Instead of predicting oral-health outcomes from social and behavioral factors, this analysis examined whether income could be predicted from dental information alone. Originally, we used DMFT as the input and trained three versions of the model: a 14-class model for NHANES household income brackets, a binary model for whether a person was above or below the federal poverty line, and a regression model for the continuous income-to-poverty ratio. However, after the inital testing results, we chose to update our model by determining the correlation between dental variables and the income-to-poverty ratio, using 40 variables ranging from the status of individual teeth. For convience, Table 2 contains a list of the variables used in the updated neural network.

Table 2: Dental variable with absolute corrlation >= 0.10 with income-to-poverty ratio.

| Variable | Correlation with Income-to-Poverty Ratio |
|---|---|
| `has_dental_implant` | 0.1159 |
| `tooth_02_status` | 0.1582 |

| Variable | Correlation with Income-to-Poverty Ratio |
|---|---|
| tooth_03_status | 0.1814 |
| tooth_04_status | 0.1731 |
| tooth_05_status | 0.1486 |
| tooth_06_status | 0.1574 |
| tooth_07_status | 0.1753 |
| tooth_08_status | 0.1629 |
| tooth_09_status | 0.1503 |
| tooth_10_status | 0.1643 |
| tooth_11_status | 0.1607 |
| tooth_12_status | 0.1423 |
| tooth_13_status | 0.1727 |
| tooth_14_status | 0.1764 |
| tooth_15_status | 0.1730 |
| tooth_18_status | 0.1744 |
| tooth_19_status | 0.1773 |
| tooth_20_status | 0.1622 |
| tooth_21_status | 0.1079 |
| tooth_22_status | 0.1524 |
| tooth_23_status | 0.1331 |
| tooth_24_status | 0.1378 |
| tooth_25_status | 0.1335 |
| tooth_26_status | 0.1347 |
| tooth_27_status | 0.1510 |
| tooth_28_status | 0.1171 |
| tooth_29_status | 0.1588 |
| tooth_30_status | 0.1751 |
| tooth_31_status | 0.1726 |
| last_dental_visit | 0.2857 |
| reason_last_dental_visit | 0.2018 |
| needed_dental_care_couldnt_get | 0.2150 |
| self_rated_oral_health | 0.2770 |
| had_oral_cancer_exam | 0.1853 |
| when_last_oral_cancer_exam | 0.1663 |
| teeth_present | 0.1573 |
| teeth_decayed | 0.1218 |
| teeth_filled_sound | 0.2238 |
| teeth_missing_caries | 0.1544 |
| untreated_decay_ratio | 0.1419 |

Networks were split into different configurations depending on the stage at which we created our neural network. Models 1, 2 and 3 had three fully connected hidden layers with 128, 64, and 32 units, ReLU activations, and a dropout rate of 0.2 between layers, while models 4, 5 and 6 had three hidden layers with 256, 128, and 64 units in order to account for the additional variables used. We trained all six model with Adam at a learning rate of 0.001, a batch size of 32, and up to 50 epochs, with early stopping based on a patience of 10 epochs on a held-out validation set. Each

version used the standard loss function for its task: categorical cross-entropy for the multi-class model, binary cross-entropy for the binary model, and mean squared error for the continuous model. Table 3 summarize the model configurations below.

Table 3: Model configurations for all six experiments. Models 1–3 use only `dmft_score` (1 feature); Models 4–6 use all 40 useful dental features.

| Model | Task | Hidden Units (h1/h2/h3) | Activation | Optimizer | Learning Rate | Dropout |
|---|---|---|---|---|---|---|
| **Model 1** | Multiclass | 128 / 64 / 32 | `relu` | Adam | 0.001 | 0.2 |
| **Model 2** | Binary | 128 / 64 / 32 | `relu` | Adam | 0.001 | 0.2 |
| **Model 3** | Regression | 128 / 64 / 32 | `relu` | Adam | 0.001 | 0.2 |
| **Model 4** | Multiclass | 256 / 128 / 64 | `relu` | Adam | 0.001 | 0.2 |
| **Model 5** | Binary | 256 / 128 / 64 | `relu` | Adam | 0.001 | 0.2 |
| **Model 6** | Regression | 256 / 128 / 64 | `relu` | Adam | 0.001 | 0.2 |

Although this deep learning track was exploratory, it played an important role in shaping the broader project. It helped us become familiar with the NHANES dental and income variables, and the early models showed that a single clinical summary such as DMFT contains some information about income on its own while models 4, 5 and 6 motivated the broader prediction framework used in the main analysis. It also provides useful context for a larger point that appears throughout the study, namely that in tabular survey data of this size, what can be learned depends not only on the model family but also on the direction of the prediction task (James et al. 2021).

All models were evaluated using 5-fold cross-validation. For the two classification tasks, we used stratified folds so that each fold preserved the overall class balance. For the regression task, we used standard random folds. Any preprocessing steps, including imputation and scaling, were fit using only the training portion of each fold and then applied to the validation portion, so that no information from the validation data leaked into training.

For the classification tasks, we report balanced accuracy, ROC-AUC, and weighted F1. For the regression task, we report MAE, RMSE, and $R^2$. We treat ROC-AUC as the main classification metric because it does not depend on a single decision threshold and remains informative under mild class imbalance. For the DMFT regression task, we treat MAE as the main metric because it is easy to interpret on the original scale of the outcome, measured in number of teeth, and because it is less affected than RMSE or $R^2$ by the large cluster of cases at `DMFT = 28`. For each model, inputs were created to allow each network to track and report a designated measure (Training Accuracy/Loss, Validation Accuracy/Loss, Training MAE, or Validation MAE).

### Two-stage model for DMFT

The DMFT outcome was not evenly distributed across values. One clear reason was a large cluster of cases at 28, which often reflected complete tooth loss. In our sample, 385 of the 608 adults with no remaining natural teeth, or 63%, had a DMFT score of exactly 28. This matters because having no teeth is a distinct state of dental health rather than a slightly more severe version of having some decay. If we fit one regression model to the entire sample, the model can become overly focused on predicting total tooth loss rather than capturing differences in DMFT among adults who still have their natural teeth. That tendency worsens prediction error and makes the results harder to interpret. To address this, we used a two-stage approach inside each cross-validation fold. In Stage 1, an XGBoost classifier predicted whether a participant was edentulous. In Stage 2, an XGBoost regressor was trained only on non-edentulous participants and used to predict their DMFT score. Anyone predicted to be edentulous in Stage 1 was assigned a DMFT value of 28, and everyone else received the continuous DMFT prediction from Stage 2. Both stages used the same hyperparameter settings as the single-stage XGBoost model.

## Results

### Sample description

The working sample included 5,265 adults. The mean age was 51.4 years (`SD = 17.7`, range 20 to 80), and 51.7% of participants were female. The self-identified race and ethnicity distribution was 34.3% Non-Hispanic White, 23.6% Non-Hispanic Black, 19.4% in the combined *Non-Hispanic Asian / Other* category (`NHANES code 5`), 13.3% Mexican American, and 9.4% Other Hispanic. This distribution reflects NHANES's intentional oversampling of minority populations.

The mean income-to-poverty ratio was 2.50 (`SD = 1.50`), and 16.5% of adults were below the federal poverty line (`IPR \leq 1`). The engineered dental measures also indicated substantial disease burden. Participants had an average of 21.9 teeth present out of a possible 32 (`SD = 10.0`), the mean DMFT score was 10.7, and 11.5% of adults were edentulous.

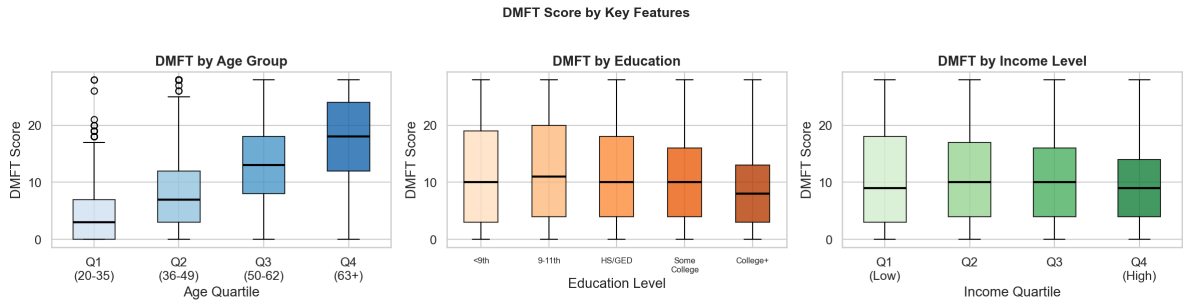


Figure 9: DMFT score stratified by age quartile, education, and income. All three panels show a clear pattern: higher disease burden among older, less educated, and lower-income adults.

The two binary targets are moderately imbalanced (34.4% and 42.5% positive, respectively), well within the range where ROC-AUC will still be reliable.

### Task 1 — Poor self-rated oral health

Table 4 reports 5-fold cross-validated performance for Task 1.

Table 4: 5-fold cross-validated performance for Task 1 (predicting *Fair* or *Poor* self-rated oral health). Tuned XGBoost used `n_estimators=1000`, `max_depth=3`, `learning_rate=0.01`, `subsample=0.8`, `colsample_bytree=0.8`, `min_child_weight=5`, `gamma=0.1`.

| Model | Balanced Accuracy | ROC-AUC | Weighted F1 |
|---|---|---|---|
| Logistic Regression | 0.755 | 0.841 | 0.768 |
| Random Forest | 0.733 | 0.843 | 0.775 |
| XGBoost (default) | 0.759 | 0.838 | 0.774 |
| **XGBoost (tuned)** | **0.770** | **0.849** | **0.785** |

All three model families converge on ROC-AUC values between 0.83 and 0.85, showing a strong and consistent signal from the input features. The tuned XGBoost has the best overall performance (ROC-AUC 0.849), but only marginally better than the regularized logistic regression with a difference of only 0.01 AUC. This is itself an important finding. A substantial share of the predictive signal in Task 1 is linear in the SES and access block of features. Feature-importance analysis ranks income-to-poverty ratio, age, time since last dental visit, reported cost-related barrier, and DMFT as the top five predictors. These five alone have about three-fourths of the total importance weight.

### Task 2 — Clinician recommendation for care

Task 2 uses the 5,057 adults with an observed examiner recommendation and a 36-feature input including self-reported symptoms.

Table 5: 5-fold cross-validated performance for Task 2 (predicting clinician-identified need for dental care). Tuned XGBoost used `n_estimators=500`, `max_depth=5`, `learning_rate=0.01`, `subsample=0.7`, `colsample_bytree=0.8`, `min_child_weight=1`, `gamma=0.1`.

| Model | Balanced Accuracy | ROC-AUC | Weighted F1 |
|---|---|---|---|
| Logistic Regression | 0.747 | 0.831 | 0.755 |
| Random Forest | 0.743 | 0.834 | 0.757 |
| XGBoost (default) | 0.747 | 0.840 | 0.755 |
| **XGBoost (tuned)** | **0.766** | **0.844** | **0.776** |

Task 2 performed slightly worse than Task 1, with AUC values of 0.83 to 0.84 compared with 0.83 to 0.85 for Task 1. This makes sense because the Task 2 outcome reflects a clinician's judgment rather than a person's own self-rating. As a result, it is somewhat less closely tied to self-reported features. Even so, the self-reported symptom variables still added useful information. In both the random forest and XGBoost models, `thinks_has_gum_disease`, `told_bone_loss_around_teeth`, and the frequency of mouth aching were all among the top fifteen features. These ranked above every care-barrier variable except cost.

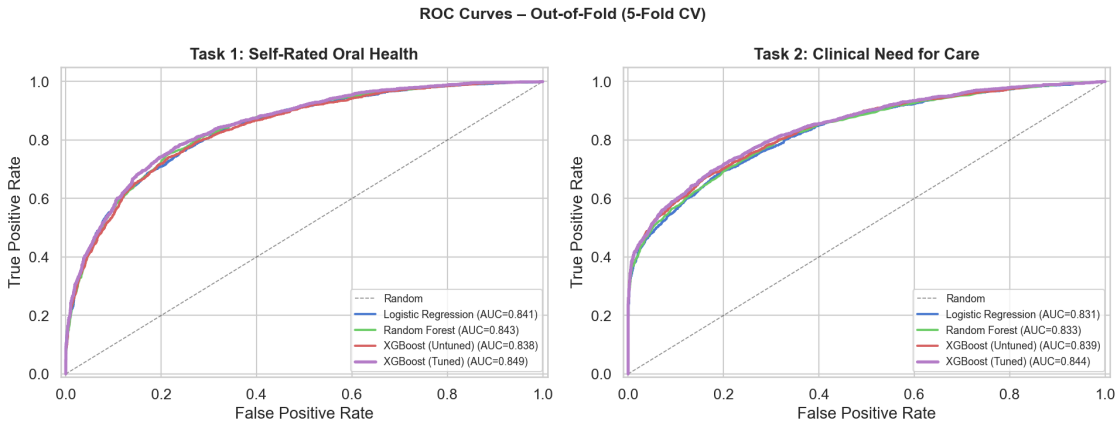


Figure 10: Out-of-fold ROC curves for Tasks 1 and 2. All four model variants cluster tightly, indicating a shared discriminative ceiling near AUC = 0.84–0.85 for both outcomes.

## Task 3 — DMFT regression

Task 3 uses 28 upstream features with all raw clinical exam variables excluded. The naive baseline (predicting the sample mean of 10.74 for every participant) yields MAE = 6.98.

Table 6: 5-fold cross-validated performance for Task 3 (predicting continuous DMFT score from upstream features only). The two-stage model classifies edentulous status in Stage 1 and assigns DMFT = 28 to those participants, regressing DMFT among the remainder in Stage 2.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Ridge Regression | 5.06 | 6.39 | 0.423 |
| Random Forest | 4.83 | 6.24 | 0.451 |
| XGBoost (single-stage) | 4.74 | 6.14 | **0.469** |
| **XGBoost (two-stage)** | **4.67** | 6.35 | 0.431 |

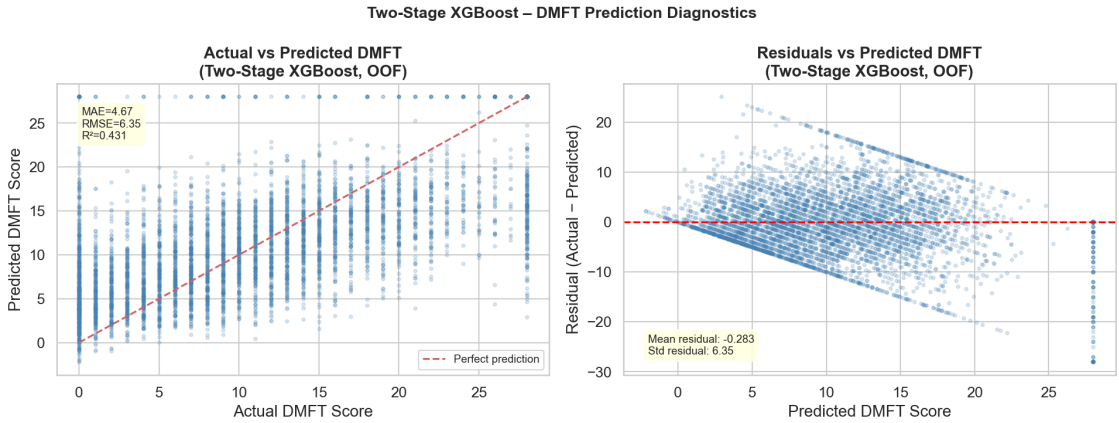


Figure 11: Actual vs. predicted DMFT and residual plot for the two-stage XGBoost (out-of-fold). The vertical band of residuals near predicted DMFT = 28 reflects cases where the Stage 1 classifier incorrectly assigns edentulous status.

All three statistical-learning models greatly outperform the naive baseline, reducing MAE by roughly 33%. The two-stage model achieves the lowest MAE (4.67) and respects the discontinuity at complete tooth loss. The single-stage XGBoost achieves a slightly higher $R^2$ (0.469 vs 0.431) because $R^2$ rewards models that capture the full DMFT distribution, including the edentu-

## Cross-task consistency

One of the clearest findings across the three tasks is how similar the most important predictors were. Income-to-poverty ratio, age, time since the last dental visit, education, and reported cost-related barriers all appeared among the top ten features for every outcome. This pattern suggests that the same broad set of social and access-related factors is shaping all three oral-health measures, which is consistent with prior public-health research on the social determinants of oral health (Braveman, Egerter, and Williams 2011; Northridge, Kumar, and Kaur 2020).

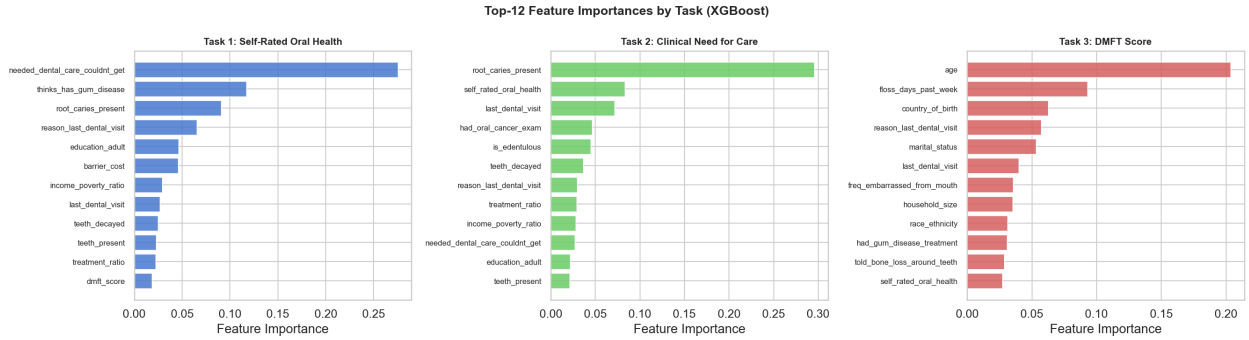


Figure 12: Top-12 XGBoost feature importances for all three tasks. The leading predictors differ by task: unmet need dominates self-rated health (Task 1), clinical severity signals dominate the examiner's recommendation (Task 2), and age dominates cumulative tooth decay (Task 3).

This also suggests that one simpler model might capture much of the signal now handled by three separate task-specific models. That approach would likely give up some accuracy for each individual task, but it could still be useful in primary care settings where a simpler screening tool is more practical.

## Neural Network

The results of the neural network models can be split into two groups: the results from only using DMFT score and the results from models that took all 40 significant variables into account. For the multiclass DMFT model, the training accuracy and validation accuracy were both less than 20%, and despite the varying range of income levels present in the data, everyone was predicted to have made over $75,000 per year regardless of their DMFT score. For the binary DMFT model, we saw similar results. Despite the data containing people who were both above and below the poverty line, we found that our model simply predicted that everyone was above the poverty line, suggesting an extremely simple model. Finally, for the linear DMFT model, we found that despite the variety of income levels present in the data, the model only predicted that everyone had an income-to-poverty ratio between 2 and 3.

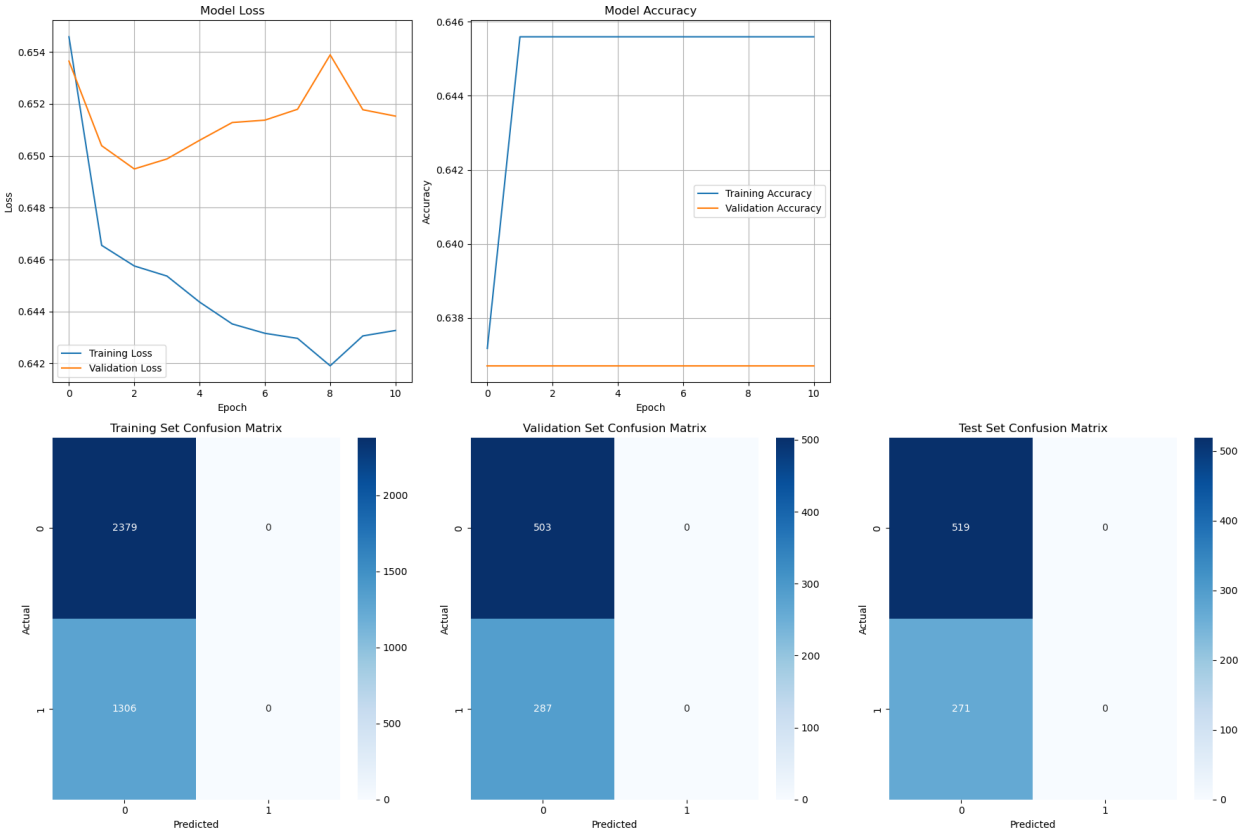


Figure 13: Binary DMFT model results. The top row tracks the loss and accuracy of the training and validation models while the bottom row shows confusion matricies of the training, testing, and validation data. '0' indicates below the poverty line while '1' indicates above the poverty line.

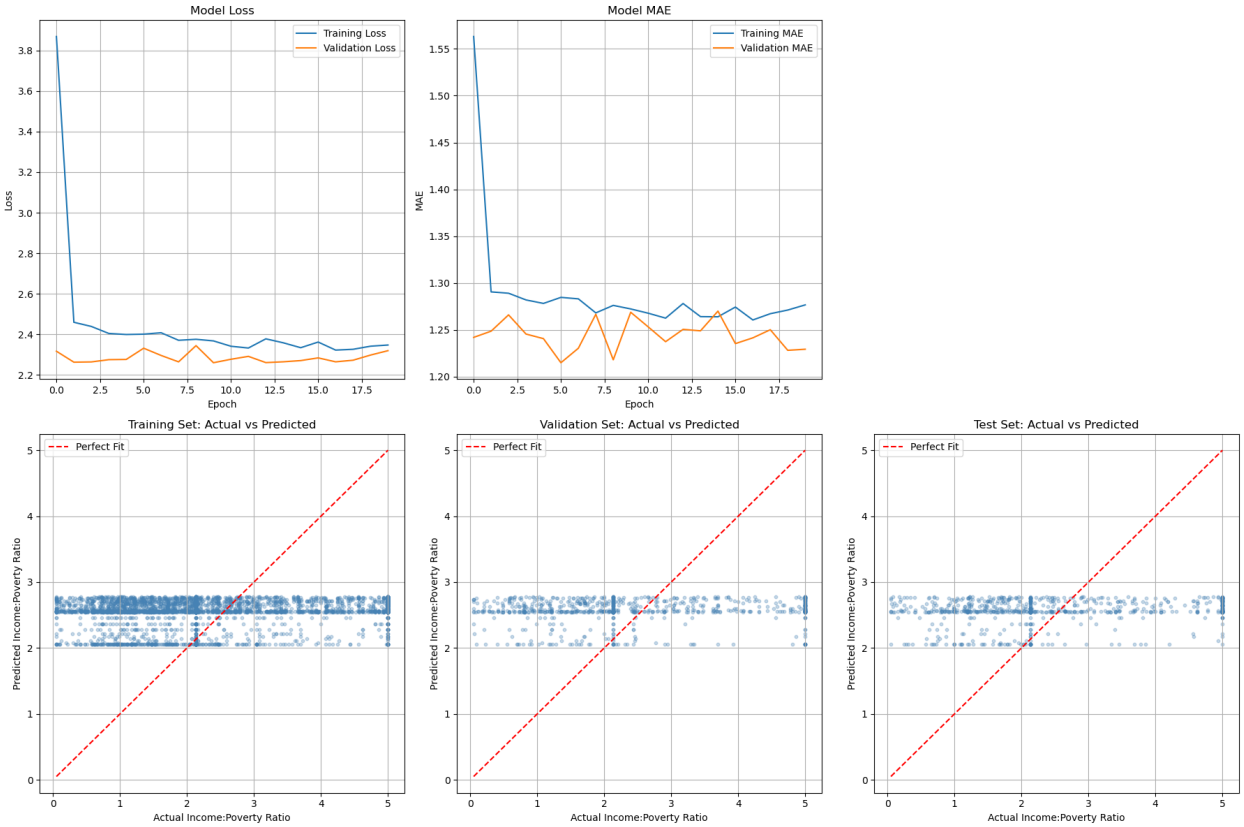


Figure 14: Regression DMFT model results. The top row tracks the loss and MAE of the training and validation models while the bottom row shows charts of the predicted vs. actual income levels of the training, testing, and validation data.

However, when the neural network models began to include the significant dental variables, our results began to appear more realistic. With the multiclass significant model, our training model reached an accuracy of around 60% with our validation accuracy reaching near 20%. For the binary classification model, our validation accuracy was over 83% and was able to identify people who were below the poverty line. Finally, our regression significant model had our training, testing and validation MAEs between 1.12 and 1.013 and provided a wider spread of predicted values.

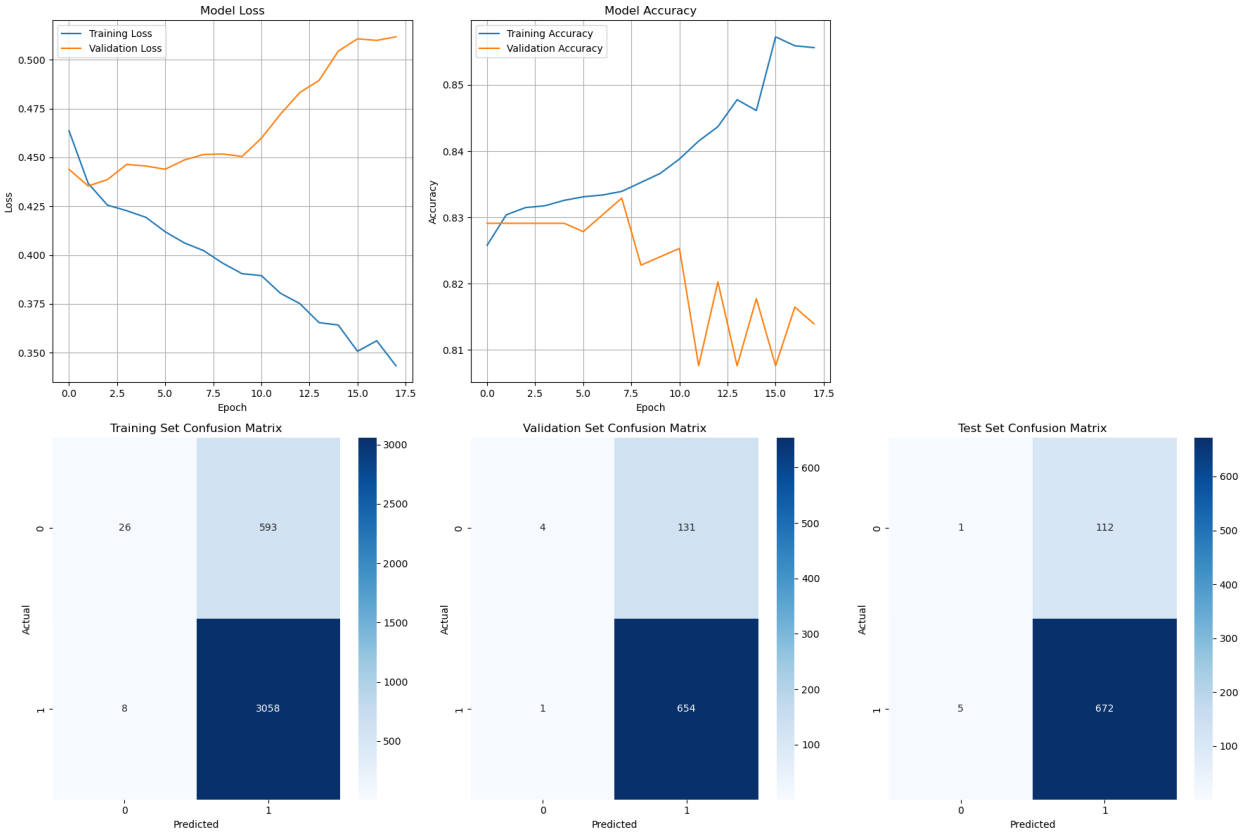


Figure 15: Binary significant model results. The top row tracks the loss and accuracy of the training and validation models while the bottom row shows confusion matricies of the training, testing, and validation data. '0' indicates below the poverty line while '1' indicates above the poverty line.

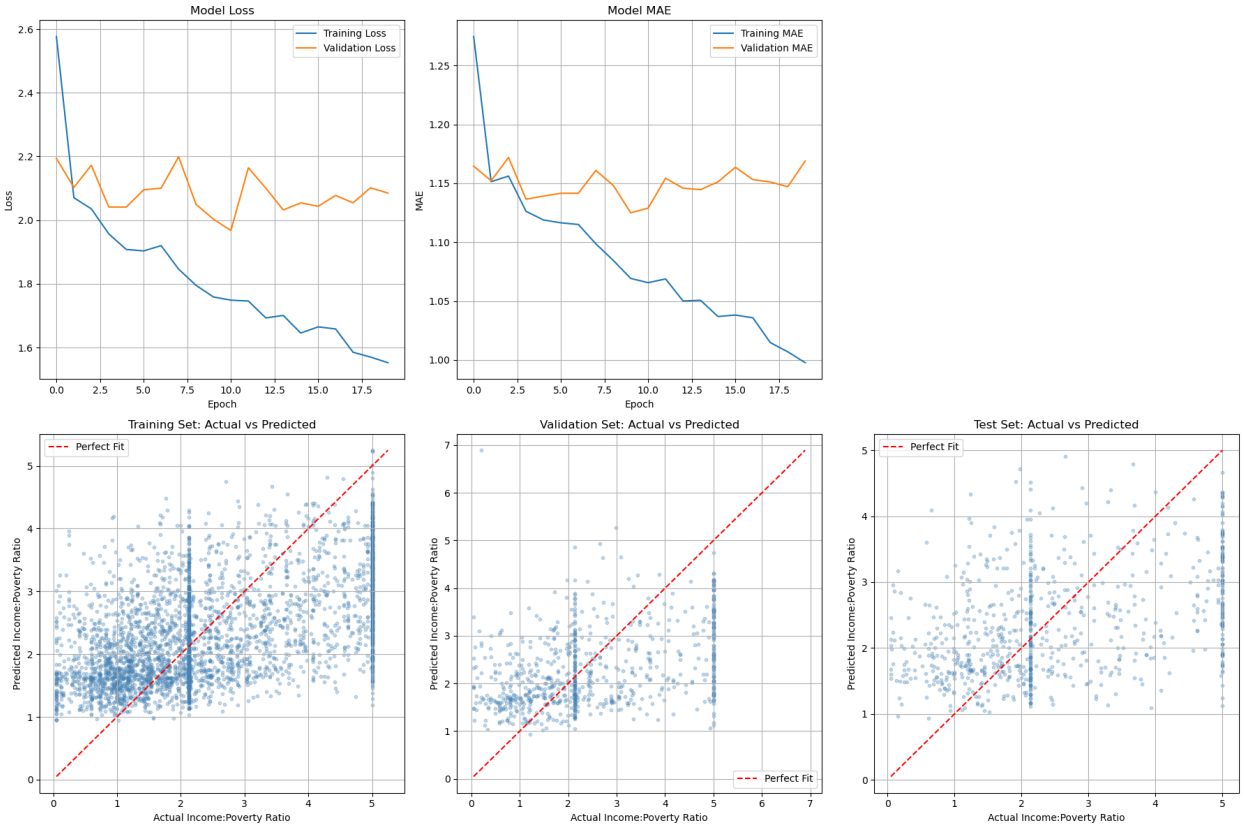


Figure 16: Regression significant model results. The top row tracks the loss and MAE of the training and validation models while the bottom row shows charts of the predicted vs. actual income levels of the training, testing, and validation data.

## Discussion

Several findings from the results are worth discussing.

**Most of the predictive signal appears to be largely linear, though some non-linearity remains.** On both binary tasks, regularized logistic regression performed within about 0.01 AUC of the tuned XGBoost model. Given the large number of socioeconomic, access, and care-barrier variables, and given that no single predictor appeared to dominate the descriptive patterns, we expected boosting to gain more from combining many weaker signals in more complex ways. Instead, the results suggest that the strongest predictors, especially income, age, education, and time since the last dental visit, operate mostly in an additive way on the log-odds scale. In practical terms, this means that a linear model with balanced class weights captures most of the available signal. The slightly stronger performance of XGBoost indicates that some additional non-linear structure is still present, likely through a limited number of interactions that the logistic model cannot represent unless they are specified directly.

**The edentulous group is important for both prediction and interpretation.** About one in nine adults in the sample had no remaining natural teeth. On average, these adults were older, had lower income, and had less education than adults who still had teeth. At the same time, they

showed a lower rate of `target_needs_care`. This pattern may appear counterintuitive, but it does not imply better oral health. A more likely explanation is that once a person has no natural teeth remaining, there are fewer tooth-level treatment recommendations an examiner can make. This creates an important modeling issue. If a model is trained on the full sample without accounting for edentulousness, it may understate care needs for one of the most vulnerable groups in the dataset.

**Limitations.** This project has three main limitations. First, the 2017 to 2018 NHANES data are cross-sectional, so all variables were measured at the same time. Because of this, we cannot say that one factor caused another. The results should be viewed as predictive, not causal. Second, we evaluated the models with standard cross-validation and did not fully account for survey weights or the full NHANES sampling design. Third, the deep learning part of the project was developed earlier, when the study focused on a different question. Because of that, it was not set up as a direct comparison with the three final tasks. Based on the shallow-model results and prior research on tabular data, a more heavily tuned neural network would probably not perform much better than gradient boosting at this sample size and feature count (James et al. 2021). Still, extending the deep learning analysis to match the final tasks would be a reasonable next step.

## Conclusion

Three findings stand out from this analysis. First, all three oral-health outcomes are predictable to a clinically meaningful degree from a relatively small set of commonly collected variables. Tuned XGBoost reached a cross-validated ROC-AUC of 0.849 for self-rated poor oral health and 0.844 for examiner-identified need for care, and the two-stage regression reduced mean absolute error on DMFT from 6.98 teeth under a naive baseline to 4.67 teeth, a 33% improvement, using only upstream predictors and no direct clinical exam data. Second, the same core set of predictors, including income-to-poverty ratio, age, time since the last dental visit, education, and cost-related barriers, appeared in the top ranks for every task. This cross-task consistency strengthens the interpretation that a common social and access-related structure underlies all three outcomes rather than each outcome being driven by its own idiosyncratic factors. Third, modeling complete tooth loss as a distinct state materially improved both predictive accuracy on the original scale and the interpretability of the results, particularly for the subgroup of adults most affected by cumulative disease burden.

Taken together, these results support two broader points. From a public-health standpoint, they reinforce the evidence that oral-health disparities in U.S. adults are tightly coupled to a small number of well-established social determinants, and that a short list of routinely collected survey items is sufficient to identify individuals at risk of poor oral health or unmet treatment need. This is relevant to screening and surveillance in primary care and community settings, where a lightweight, non-clinical instrument could help flag adults who would benefit from targeted outreach or help finding care, even in the absence of a dental exam. From a statistical learning standpoint, the near-equivalence between regularized logistic regression and tuned gradient boosting on the two classification tasks is a good reminder that on moderately sized tabular survey data with strong, largely additive predictors, flexible non-linear methods offer incremental gains.

## References


Braveman, Paula, Susan Egerter, and David R. Williams. 2011. "The Social Determinants of Health: Coming of Age." *Annual Review of Public Health* 32: 381–98.

Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32.

Centers for Disease Control and Prevention, National Center for Health Statistics. 2020. "National Health and Nutrition Examination Survey, 2017–2018: Demographics, Oral Health Dentition, Oral Health Questionnaire, and Oral Health Recommendation of Care Files." https://wwwn.cdc.gov/Nchs/Nhanes/2017-2018/.

Dye, Bruce A., Darien J. Weatherspoon, and Gabriela Lopez Mitnik. 2019. "Tooth Loss Among Older Adults According to Poverty Status in the United States from 1999 Through 2004 and 2009 Through 2014." *The Journal of the American Dental Association* 150 (1): 9–23.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning with Applications in R*. 2nd ed. Springer.

Klein, Henry, Carroll E. Palmer, and John W. Knutson. 1938. "Studies on Dental Caries. I. Dental Status and Dental Needs of Elementary School Children." *Public Health Reports* 53 (19): 751–65.

Locker, David, and Carlos Quinonez. 2009. "Functional and Psychosocial Impacts of Oral Disorders in Canadian Adults: A National Population Survey." *Journal of the Canadian Dental Association* 75 (7): 521.

Northridge, Mary E., Anjali Kumar, and Rajvir Kaur. 2020. "Disparities in Access to Oral Health Care." *Annual Review of Public Health* 41: 513–35.

Peres, Marco A., Lorna M. D. Macpherson, Robert J. Weyant, Blánaid Daly, Renato Venturelli, Manu R. Mathur, Stefan Listl, et al. 2019. "Oral Diseases: A Global Public Health Challenge." *The Lancet* 394 (10194): 249–60. https://doi.org/10.1016/S0140-6736(19)31146-8.

Sabbah, Wael, Georgios Tsakos, Tarani Chandola, Aubrey Sheiham, and Richard G. Watt. 2007. "Social Gradients in Oral and General Health." *Journal of Dental Research* 86 (10): 992–96.

Slade, Gary D. 1997. "Derivation and Validation of a Short-Form Oral Health Impact Profile." *Community Dentistry and Oral Epidemiology* 25 (4): 284–90.

World Health Organization. 2025. "Oral Health." https://www.who.int/news-room/fact-sheets/detail/oral-health.